



Procurement and Property Management

Office of the Chief Financial Officer

1 Cyclotron Rd. | MS: 71-R0259 | Berkeley, CA 94720

February 27, 2026

Subject: **Request for Information (RFI) No. AA-2026-02**

Response Due Date: March 9, 2026

Introduction

The University of California, Lawrence Berkeley National Laboratory (LBNL) is conducting market research to identify potential sources that possess the expertise, capabilities, and experience necessary to supply the Beam Position Monitor (BPM) major and sub-assemblies for the Advanced Light Source (ALS-U) project.

Background

ALS-U is an upgrade project to the LBNL Advanced Light Source (ALS). The ALS is a 1.9 GeV storage ring operating at 500 mA of beam current. It is optimized to produce intense beams of soft x-rays, which offer spectroscopic contrast, nanometer-scale resolution, and broad temporal sensitivity. The ALS facility includes an accelerator complex and photon delivery system that is capable of providing the foundations for an upgrade that will achieve world-leading soft x-ray coherent flux. The existing ALS provides a ready-made foundation, including conventional facilities and also includes extensive beamlines (up to 40 simultaneously operating beamlines) and instrumentation, an experimental hall, computing resources, ancillary laboratories, offices, and related infrastructure that will be heavily utilized in an upgrade scenario.

The ALS will be upgraded with a multibend achromat (MBA) lattice design that will provide a soft x-ray source that is orders of magnitudes brighter — an up to 100-1000 times increase in brightness and will provide a significantly higher fraction of coherent light in the soft x-ray region than is currently available at ALS.

Use Case

BPM electronics assemblies provide signal conditioning, timing distribution, and processing necessary to measure the transverse position of the particle beam. The assemblies consist of multiple printed circuit board (PCB) sub-assemblies grouped into major functional assemblies, including the Timing + Balun assembly, Analog Front End (AFE) assemblies, and Chassis-level electronics.

Scope

LBNL is conducting market research to identify potential sources with the capability to supply the BPM major and sub-assemblies; that can demonstrate the following requirements:

1. PCB Design / Layout Capabilities

- Indicate whether PCB layout is performed in-house or by a subcontractor
- List the CAD tools utilized (Altium, Cadence, Mentor, etc.)
- Describe experience in the following design areas:
 - High-speed digital
 - RF / analog
 - Mixed signal
 - High-voltage
 - Controlled impedance
- Describe stack-up capabilities (layer count limits, blind/buried vias, microvias)

- Define applicable DFM/DFT process and whether or not formal design reviews are conducted.
- Confirm capability and provide specifics of:
 - Signal Integrity (SI) analysis
 - Power Integrity (PI) analysis
 - Thermal analysis
- Describe design documentation standards and IPC compliance

2. PCB Fabrication Capabilities

- Provide fabrication limits and capabilities for the following:
 - Layer count (max/min)
 - Trace width / spacing (max/min)
 - Via size (max/min)
- What is your controlled impedance tolerance?
- Please state what material types are supported (FR-4, Rogers, polyimide, etc.).
- Please provide your current lead times for job requests.

3. PCB Assembly

Please provide your expertise in the following areas:

- SMT and through-hole capabilities?
- Fine-pitch BGA capability? (What pitch?)
- X-ray inspection available?
- AOI available?
- Rework capability (BGA reballing?)
- Conformal coating?
- Functional test capability?
- Burn-in capability?
- Cleanliness standards (IPC Class 2 or 3?)
- Lot traceability system?
- ESD controls?

4. Testing & Quality

Please provide details on your testing capabilities and quality standards such as the examples below:

- ISO certification (ISO 9001, AS9100, etc.)
- IPC certifications

5. Chassis & Integration Capabilities

- Do you do mechanical fabrication in-house?
- Sheet metal? CNC? Powder coating?
- Cable harness fabrication?
- Rack integration capability?
- System-level wiring?



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- High-voltage wiring experience?
- Thermal management design?
- Shock/vibration considerations?
- Can you perform full system bring-up and test?

6. Programmatic & Capacity Questions

- Please provide your current production capacity.
- What are your main lead time drivers and supply chain constraints?

7. General

- Can you please provide a summarized capability matrix
- Please provide at least 3 similar project examples

Instructions

Respondents who can demonstrate a proven record of the identified scope should email the LBNL Procurement Representative Arielle Anoop at AAnoop@lbl.gov with a list of previous projects and any published product literature and/or technical specifications. Responses must be submitted by email no later than 3:00pm PT on March 9, 2026.

NAICS Code And Size Standard

The North American Industry Classification System (NAICS) Code for this RFI is 334418 (Printed Circuit Assembly – Electronic Assembly Manufacturing). The corresponding small business size standard for this is 750 employees.

Request for Information Only

This RFI is for informational and planning purposes only. This is not a request for quote or proposal. It does not constitute a solicitation and shall not be construed as a commitment by the University. Responses in any form are not offers and no subcontract will be awarded as a result of this announcement. No inquiries regarding a bid package or solicitation will be answered, because there is no bid package or solicitation available. Any information submitted by respondents to this RFI is strictly voluntary.

In order to protect the procurement integrity of any future procurement that may arise from this RFI, information regarding the technical point of contact will not be given and no appointments for presentations will be made. No funds are available to pay for preparation of responses to this announcement, and no reimbursement will be made for any costs associated with providing information in response to this RFI or any follow-up information requests related to this RFI. Responding to this RFI does not provide any advantage or disadvantage to potential offerors if LBNL chooses to issue a formal solicitation at a later date.

This Request for Information is governed by procurement policies and procedures established under the University's Prime Contract with the U.S. Government, represented by the Department of Energy ("DOE"), for management and operation of LBNL.

Proprietary Information

Because information received in response to this RFI may be used to structure future programs and may be made available to the public, respondents must not include any information in their responses that might be



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considered business sensitive, proprietary, or otherwise confidential. Responses must be submitted with the understanding that their contents may be publicly disclosed and, in the event of a public disclosure, LBNL will not notify respondents or provide any opportunity to revise or redact submitted information.

Arielle Anoop
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(End of RFI document)